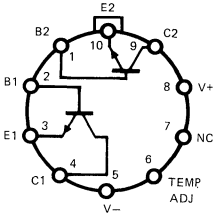
μA726

TEMPERATURE-CONTROLLED DIFFERENTIAL PAIR

FAIRCHILD LINEAR INTEGRATED CIRCUITS

GENERAL DESCRIPTION – The μA726 is a Monolithic Transistor Pair in a high thermal-resistance package, held at a constant temperature by active temperature regulator circuitry. The transistor pair displays the excellent matching, close thermal coupling and fast thermal response inherent in monolithic construction. The high gain and low standby dissipation of the regulator circuit permits tight temperature control over a wide range of ambient temperatures. It is intended for use as an input stage in very-low-drift dc amplifiers, replacing complex chopper-stabilized amplifiers. It is also useful as the nonlinear element in logarithmic amplifiers and multipliers where the highly predictable exponential relation between emitter-base voltage and collector current is employed. The device is constructed on a single silicon chip using the Fairchild Planar* process.

CONNECTION DIAGRAM
10-PIN METAL CAN
(TOP VIEW)
PACKAGE OUTLINE 5U
PACKAGE CODE H



ABSOLUTE MAXIMUM RATINGS

Operating Temperature Range	
Military (μA726)	–55°C to +125°C
Commercial (μA726C)	0°C to +85°C
Storage Temperature Range	–65°C to +150°C
Pin Temperature (Soldering, 60 seconds)	300°C
Supply Voltage	±18V
Internal Power Dissipation	500mW

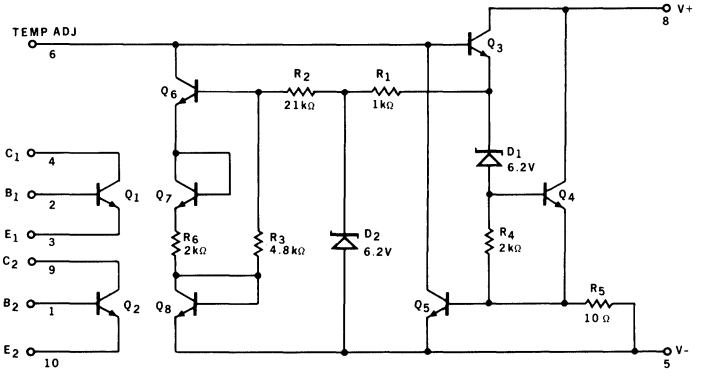
MAXIMUM RATINGS FOR EACH TRANSISTOR

Collector-to-Emitter Voltage, V_{CEO}	30V
Collector-to-Base Voltage, V_{CBO}	40V
Collector-to-Substrate Voltage, V_{CIO}	40V
Emitter-to-Base Voltage, V_{EBO}	5V
Collector Current, I_C	5mA

ORDER INFORMATION

TYPE	PART NO.
μA726	μA726HM
μA726C	μA726HC

EQUIVALENT CIRCUIT



*Planar is a patented Fairchild process.

μA726

ELECTRICAL CHARACTERISTICS: $-55°C \leq T_A \leq +125°C$, $V_S = \pm 15V$, $R_{adj} = 62k\Omega$ unless otherwise specified.

CHARACTERISTICS	CONDITIONS	MIN	TYP	MAX	UNITS
Input Offset Voltage	$10\mu A \leq I_C \leq 100\mu A$, $V_{CE} = 5V$, $R_S \leq 50\Omega$		1.0	2.5	mV
Input Offset Current	$I_C = 10\mu A$, $V_{CE} = 5V$		10	50	nA
	$I_C = 100\mu A$, $V_{CE} = 5V$		50	200	nA
Average Input Bias Current	$I_C = 10\mu A$, $V_{CE} = 5V$		50	150	nA
	$I_C = 100\mu A$, $V_{CE} = 5V$		250	500	nA
Offset Voltage Change	$I_C = 10\mu A$, $5V \leq V_{CE} \leq 25V$, $R_S \leq 100k\Omega$		0.3	6.0	mV
	$I_C = 100\mu A$, $5V \leq V_{CE} \leq 25V$, $R_S \leq 10k\Omega$		0.3	6.0	mV
Input Offset Voltage Drift	$10\mu A \leq I_C \leq 100\mu A$, $V_{CE} = 5V$, $R_S \leq 50\Omega$, $+25°C \leq T_A \leq +125°C$		0.2	1.0	μV/°C
Input Offset Voltage Drift	$10\mu A \leq I_C \leq 100\mu A$, $V_{CE} = 5V$, $R_S \leq 50\Omega$, $-55°C \leq T_A \leq +25°C$		0.2	1.0	μV/°C
Input Offset Current Drift	$I_C = 10\mu A$, $V_{CE} = 5V$		10		pA/°C
	$I_C = 100\mu A$, $V_{CE} = 5V$		30		pA/°C
Supply Voltage Rejection Ratio	$10\mu A \leq I_C \leq 100\mu A$, $R_S \leq 50\Omega$,		25		μV/V
Low Frequency Noise	$I_C = 10\mu A$, $V_{CE} = 5V$, $R_S \leq 50\Omega$ BW = .001 Hz to 0.1 Hz		4.0		μV p-p
Broadband Noise	$I_C = 10\mu A$, $V_{CE} = 5V$, $R_S \leq 50\Omega$ BW = 0.1 Hz to 10kHz		10		μV p-p
Long-term Drift	$10\mu A \leq I_C \leq 100\mu A$, $V_{CE} = 5V$, $R_S \leq 50\Omega$, $T_A = 25°C$		5.0		μV/week
High Frequency Current Gain	f = 20MHz, $I_C = 100\mu A$, $V_{CE} = 5V$	1.5	3.5		
Output Capacitance	$I_E = 0$, $V_{CB} = 5V$		1.0		pF
Emitter Transition Capacitance	$I_E = 100\mu A$		1.0		pF
Collector Saturation Voltage	$I_B = 100\mu A$, $I_C = 1mA$		0.5	1.0	V

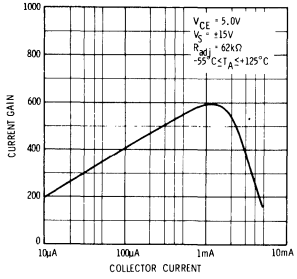
μA726C

ELECTRICAL CHARACTERISTICS: $0°C \leq T_A \leq +85°C$, $V_S = \pm 15V$, $R_{adj} = 75k\Omega$ unless otherwise specified.

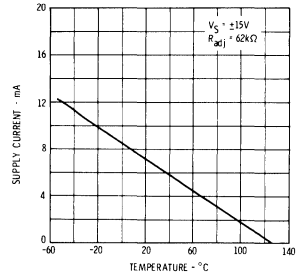
CHARACTERISTICS	CONDITIONS	MIN	TYP	MAX	UNITS
Input Offset Voltage	$10\mu A \leq I_C \leq 100\mu A$, $V_{CE} = 5V$, $R_S \leq 50\Omega$		1.0	3.0	mV
Input Offset Current	$I_C = 10\mu A$, $V_{CE} = 5V$		10	100	nA
	$I_C = 100\mu A$, $V_{CE} = 5V$		50	400	nA
Average Input Bias Current	$I_C = 10\mu A$, $V_{CE} = 5V$		50	300	nA
	$I_C = 100\mu A$, $V_{CE} = 5V$		250	1000	nA
Offset Voltage Change	$I_C = 10\mu A$, $5V \leq V_{CE} \leq 25V$, $R_S \leq 100k\Omega$		0.3	6.0	mV
	$I_C = 100\mu A$, $5V \leq V_{CE} \leq 25V$, $R_S \leq 10k\Omega$		0.3	6.0	mV
Input Offset Voltage Drift	$I_C = 100\mu A$, $V_{CE} = 5V$, $R_S \leq 50\Omega$		0.2	2.0	μV/°C
Input Offset Current Drift	$I_C = 10\mu A$, $V_{CE} = 5V$		10		pA/°C
	$I_C = 100\mu A$, $V_{CE} = 5V$		30		pA/°C
Supply Voltage Rejection Ratio	$I_C = 100\mu A$, $R_S = 50\Omega$		25		μV/V
Low Frequency Noise	$I_C = 10\mu A$, $V_{CE} = 5V$, $R_S \leq 50\Omega$, BW = 0.001 Hz to 0.1 Hz		4.0		μV p-p
Broadband Noise	$I_C = 10\mu A$, $V_{CE} = 5V$, $R_S \leq 50\Omega$, BW = 0.1 Hz to 10kHz		10		μV p-p
Long-Term Drift	$I_C = 100\mu A$, $V_{CE} = 5V$, $R_S \leq 50\Omega$, $T_A = 25°C$		5.0		μV/week
High Frequency Current Gain	f = 20MHz, $I_C = 100\mu A$, $V_{CE} = 5V$	1.5	3.5		
Output Capacitance	$I_E = 0$, $V_{CB} = 5V$		1.0		pF
Emitter Transition Capacitance	$I_E = 100\mu A$		1.0		pF
Collector Saturation Voltage	$I_B = 100\mu A$, $I_C = 1mA$		0.5	1.0	V

TYPICAL PERFORMANCE CURVES FOR μA726

CURRENT GAIN AS A FUNCTION OF COLLECTOR CURRENT

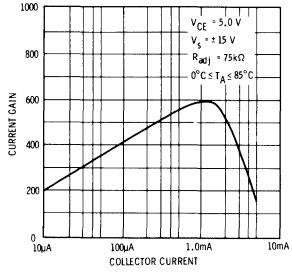


SUPPLY CURRENT AS A FUNCTION OF AMBIENT TEMPERATURE

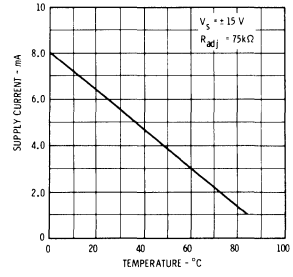


TYPICAL PERFORMANCE CURVES FOR μA726C

CURRENT GAIN AS A FUNCTION OF COLLECTOR CURRENT



SUPPLY CURRENT AS A FUNCTION OF AMBIENT TEMPERATURE



7

TYPICAL X1000 AMPLIFIER CIRCUIT

